

LAB TASKS WEEK 8

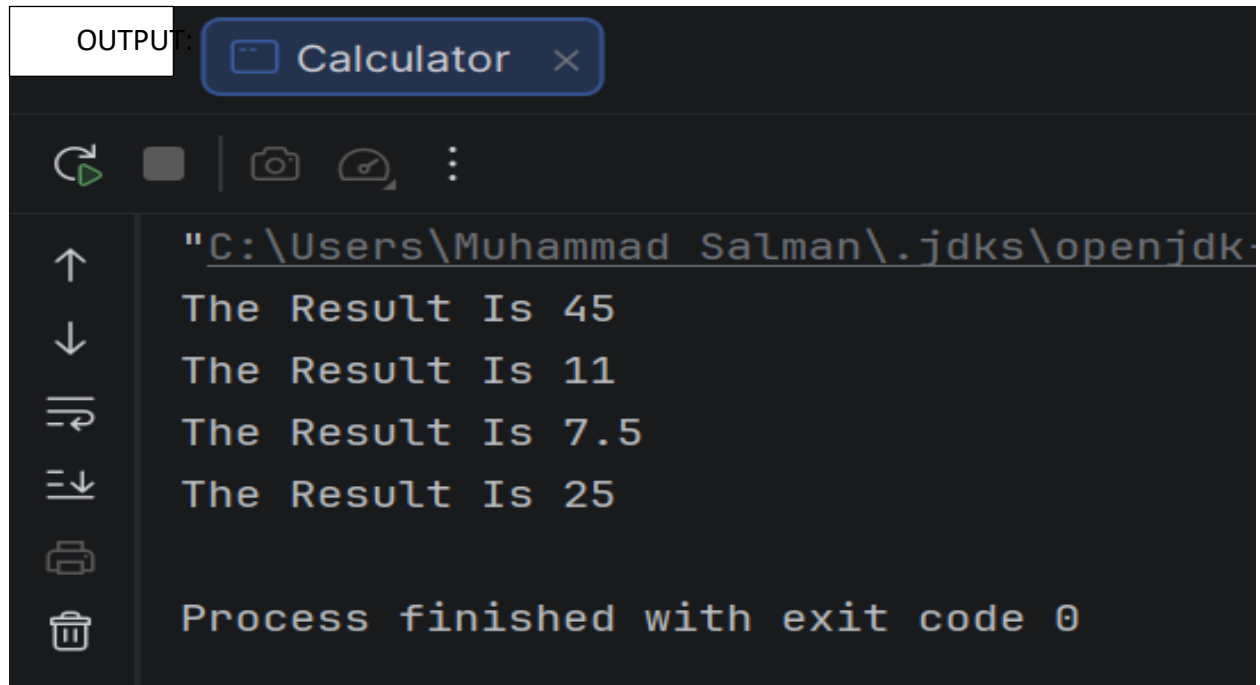
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TASK 1:

POLYMORPHISM

```
class Calculator {
    void sum() {
        int a, b, c;
        a = 5; b = 6;
        c = a + b;
        System.out.println("The Results is " + c);
    }
    void sum(int a, int b) {
        int c = a + b;
        System.out.println("The Results is " + c);
    }
    void sum(double a, int b) {
        double c = a + b;
        System.out.println("The Results is " + c);
    }
    void sum(int a, int b, int c) {
        int res = a + b + c;
        System.out.println("The Results is " + res);
    }
    public static void main(String[] args) {
        int a, b, c;
        a = 10;
        b = 15;
        c = 20;
        Calculator d = new Calculator();
        d.sum(a, b, c);
        d.sum(); d.sum(
            2.5, 5);
        d.sum(a, b);
    }
}
```

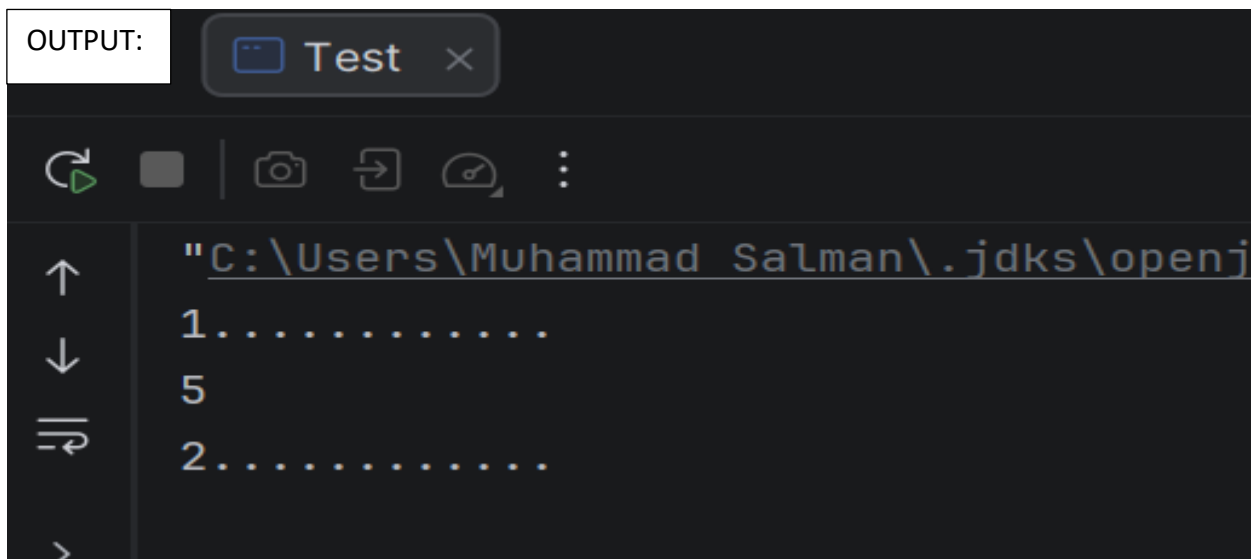


```
OUTPUT: Calculator x
" C:\Users\Muhammad Salman\.jdk\openjdk-
The Result Is 45
The Result Is 11
The Result Is 7.5
The Result Is 25
Process finished with exit code 0
```

TASK 2:**POLYMORPHISM TEST**

```
class Test{  
    void show(int a)  
        {System.out.println("1.....");  
        System.out.println(a);  
    }  
    void show()  
        {System.out.println("2.....");  
    }  
    public static void main(String[] args){ Test  
        t=new Test();  
        t.show(5);  
        t.show();  
    }  
}
```

OUTPUT:



```
"C:\Users\Muhammad Salman\.jdk\openj  
1.....  
5  
2.....
```

TASK3:**SUBTRACT THROUGH POLYMORPHISM**

```

class Calculator {
    void sub(int a, int b)
    { System.out.println(a-
      b);
    }
    void sub()
    { int a=10;
      int b = 5;
      int c;
      c=a-b;
      System.out.println(c);
    }
    void sub(double a, int b)
    { double c=a-
      b; System.out.println(c);
    }
    void sub(int a, int b, int c)
    { int res=a-b-
      c; System.out.println(res);
    }
    public static void main(String[] args) { int
      a,b,c;
      a=10;
      b=8;
      c=5;
      Calculator d=new Calculator();
      d.sub();
      d.sub(a,b);
      d.sub(5.5,3);
      d.sub(a,b,c);
    }
}

```

OUTPUT:

```
Calculator x
```

↺ | 📷 | 🗨 | ⋮

↑ "C:\Users\Muhammad Salman\.jdk\openjdk
5
↓ 2
⇌ 2.5
⇓ -3
🖨
🗑 Process finished with exit code 0

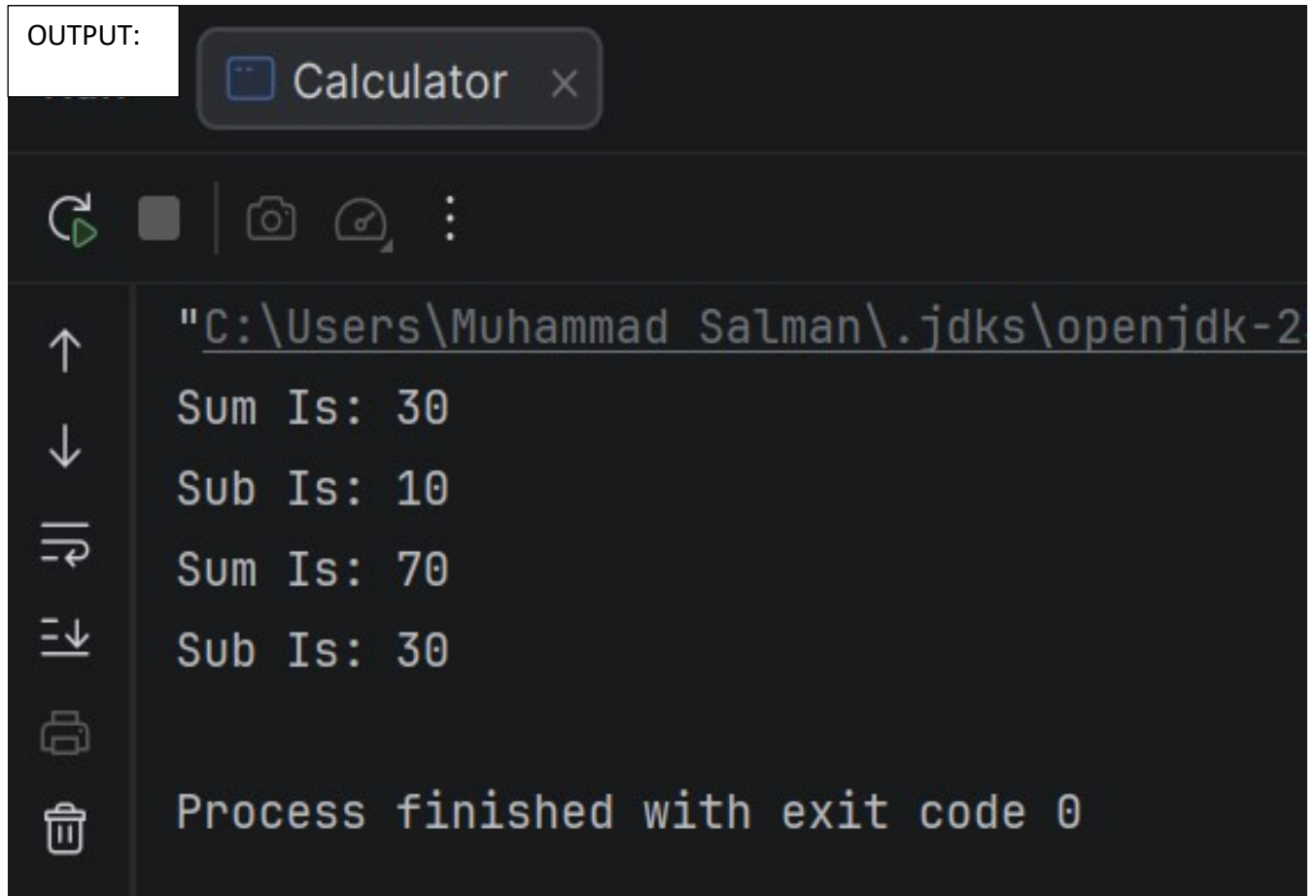
TASK4:**POLYMORPHISM SUBTRACT AND ADD**

```

class Calculator{
    void sum(){
        int a=10;
        int b=20;
        int c=a+b;
        System.out.println("SumIs:"+c);
    }
    void sub()
    {
        int
        a=20; int
        b=10;
        int c=a-b;
        System.out.println("SubIs:"+c);
    }
    void sum(inta,intb)
    {
        int c=a+b;
        System.out.println("SumIs:"+c);
    }
    void sub(inta,intb){
        int
        c=a-b;
        System.out.println("SubIs:"+c);
    }
    public static void main(String[] args){
        int
        a=50;
        int b=20;
        Calculator c=new Calculator();
        c.sum();
        c.sub();
        c.sum(a,b);
        c.sub(a,b);
    }
}

```

OUTPUT:



The screenshot shows a Windows Command Prompt window with the title bar 'Calculator'. The command prompt displays the following output:

```
"C:\Users\Muhammad Salman\.jdk\openjdk-2  
Sum Is: 30  
Sub Is: 10  
Sum Is: 70  
Sub Is: 30  
  
Process finished with exit code 0
```

The output is displayed in a monospaced font. The first line is the path to the Java JDK, followed by four lines of arithmetic results. The final line indicates that the process finished successfully with exit code 0.